Well Ordering and the Induction Principle

- **Definition 1**
  Let $A \subseteq \mathbb{Z}$. We say that A is **bounded below** if there exists $l \in \mathbb{Z}$ such that $l \leqslant a$ for all $a \in A$ and l is called a **lower bound** of A. If l is a lower bound of A and $l \in A$, we say that l is the **first element** of A (minimum).
- **Remark 1**
  The first element of A, if it exists, is unique.
  If l and l' are both first elements of A, then $l \leqslant l'$ and $l' \leqslant l$, therefore $l = l'$.

- **Definition 2**
  Let $A \subseteq \mathbb{Z}$. We say that A is **bounded above** if there exists $u \in \mathbb{Z}$ such that $a \leqslant u$ for all $a \in A$ and u is called an **upper bound** of A. If u is an upper bound of A and $u \in A$, we say that u is the **last element** of A (maximum).
- **Remark 2**
  The last element of A, if it exists, is unique.
  If u and u' are both last elements of A, then $u' \leqslant u$ and $u \leqslant u'$, therefore $u = u'$.
  - **Example 2.1**
    1. The set $A = \{a \in \mathbb{Z} : a \geqslant -2\}$ is bounded below but is not bounded above.
       $$A = \{-2, -1, 0, 1, 2, \cdots\}.$$
       Its first element is -2. Note that -3 is also a lower bound of A but $-3 \notin A$.
    2. The set $B = \{a \in \mathbb{Z} : a \leqslant 1\}$ is bounded above but is not bounded below.
       $$B = \{\cdots, -3, -2, -1, 0, 1\}.$$
       Its last element is 1. Note that 2 is also an upper bound of B but $2 \notin B$.
    3. The set $C = \{a \in \mathbb{Z} : a > 10\}$ is bounded below but is not bounded above.
       $$C = \{11, 12, 13, 14, 15, \cdots\}.$$
       Its first element is 11. Note that 10 is also a lower bound of C but $10 \notin C$.
    4. The set $\mathbb{N}$ is bounded below but is not bounded above.
       $$\mathbb{N} = \{1, 2, 3, 4, 5, \cdots\}.$$
       Its first element is 1. Note that 0 is also a lower bound of $\mathbb{N}$ but $0 \notin \mathbb{N}$. $\mathbb{N}$ is not bounded above since if $n \in \mathbb{N}$ then $n + 1 \in \mathbb{N}$ with $n < n + 1$. Therefore it is not possible to find $k \in \mathbb{N}$ such that $n \leqslant k$ for all $n \in \mathbb{N}$.

  Every subset of $\mathbb{N}$ is bounded below at least by 1.

  Well ordering axiom: Every nonempty subset of $\mathbb{N}$ has a first element.

  - **Theorem 2.2**
    Every nonempty subset $A \subseteq \mathbb{N}$ which is bounded above in $\mathbb{N}$ has a last element.
    - **Proof 2.2.1**
      Since A is bounded above in $\mathbb{N}$, let $y \in \mathbb{N}$ be such an upper bound of A.
      $$U = \{u \in \mathbb{N} : u \text{ is an upper bound of } A\}.$$
      Then $U \subseteq \mathbb{N}$ and $U \neq \varnothing$ because $y \in U$, so by the well ordering axiom U has a first element n. Note that $n - 1 \notin U$ since if $n - 1 \in U$ then $n \leqslant n - 1$ because n is a lower bound of U, which is not possible.

Therefore there exists $a \in A$ such that $n-1 < a$ and $a \leqslant n$ because n is an upper bound of A. Then $a = n$ because there is no natural number between $n-1$ and n. Therefore n is the last element of A.

- **Proposition 2.3**
  Every nonempty subset $A \subseteq \mathbb{Z}$ that is bounded below has a first element.
  - **Proof 2.3.1**
    Let $A \subseteq \mathbb{Z}$ be bounded below with $A \neq \varnothing$, y be a lower bound of A and consider the set

    $$B = \{a - y + 1 : a \in A\}.$$

    Then $B \subseteq \mathbb{N}$ and $B \neq \varnothing$ because $y \leqslant a$ for all $a \in A$, that is, $a - y \geqslant 0$ for all $a \in A$ and thus $a - y + 1 \geqslant 1$ for all $a \in A$, therefore $B \subseteq \mathbb{N}$ and $B \neq \varnothing$ because $A \neq \varnothing$. Then by the well ordering axiom B has a first element b_0, that is, $b_0 = a_0 - y + 1$ for some $a_0 \in A$, hence

    $$\begin{aligned} b_0 &\leqslant a - y + 1 && (\forall a \in A) \\ a_0 - y + 1 &\leqslant a - y + 1 && (\forall a \in A) \\ a_0 &\leqslant a. && (\forall a \in A) \end{aligned}$$

    Therefore a_0 is the first element of A.

- **Theorem 3 (The principle of induction)**
  Let $S \subseteq \mathbb{N}$ that satisfies the following conditions:
  1. $1 \in S$.
  2. If $n \in S$ then $n+1 \in S$.

  Then $S = \mathbb{N}$.
  - **Proof 3.1**
    Let $S' = \mathbb{N} \setminus S$ and suppose that $S \neq \mathbb{N}$. Then $S' \neq \varnothing$ and by the well ordering axiom S' has a first element m. Since $1 \in S$, then $m > 1$ and thus $m - 1 \in \mathbb{N}$. Since m is the first element of S' then $m - 1 \in S$. And by $\text{condition } (2)$, we have that $m = (m-1) + 1 \in S$ which is a contradiction because $m \in S'$. Therefore $S' = \varnothing$ and $S = \mathbb{N}$.
- **Corollary 3 (The principle of induction, alternative form)**
  Let $P(n)$ be a statement defined for every $n \in \mathbb{N}$. Suppose that:
  1. $P(1)$ is true.
  2. If $P(n)$ is true then $P(n+1)$ is true.

  Then $P(n)$ is true for every $n \in \mathbb{N}$.

  The $\text{condition } (1)$ is called the **base step**, the assumption that $P(n)$ is true is called the **inductive hypothesis** and the passage from $P(n)$ to $P(n+1)$ is called the **inductive step**.
  - **Example 3.1**
    Prove that $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$ for every $n \in \mathbb{N}$.
    - **Proof 3.1.1**
      1. Base step
         If $n = 1$ the result is true since $1 = \dfrac{1(1+1)}{2}$.
      2. Inductive step
         Suppose that the result is true for n, that is

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Hence

$$\begin{aligned}
1 + 2 + \cdots + n + (n+1) &= (1 + 2 + \cdots + n) + (n+1) \\
&= \frac{n(n+1)}{2} + (n+1) \\
&= \frac{n(n+1) + 2(n+1)}{2} \\
&= \frac{(n+2)(n+1)}{2}.
\end{aligned}$$

Then the result is true for $n+1$ and therefore

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2},$$

is true for every $n \in \mathbb{N}$.

- **Example 3.2**
  Prove that $n! \geqslant 2^{n-1}$ for every $n \in \mathbb{N}$.
  - **Proof 3.2.1**
    1. Base step
       If $n = 1$ the result is true since $1! = 1 \geqslant 1 = 2^{1-1}$.
    2. Inductive step
       Suppose that the result is true for n, that is

       $$n! \geqslant 2^{n-1}.$$

       Because $n \geqslant 1$, then $n + 1 \geqslant 2$, hence

       $$(n+1)! = n! \cdot (n+1) \geqslant 2^{n-1} \cdot 2 = 2^n.$$

       Then the result is true for $n+1$ and therefore

       $$n! \geqslant 2^{n-1},$$

       is true for every $n \in \mathbb{N}$.